TOOLSHARE

Project Documentation — Phases 1–3

Project Planning & Management | Literature Review | Requirements Gathering

DEPI Industry Project | October–December 2026

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Project Overview

ToolShare is a proposed peer-to-peer web platform that helps people rent tools and equipment from other people instead of buying items they may only need occasionally. Tool owners can publish listings and manage booking requests, while renters can discover tools, review details, and request rentals for selected dates.

Problem Statement

Many tools are expensive and used only occasionally. At the same time, people may own tools that remain unused for long periods. Finding a suitable tool nearby, confirming availability, and coordinating a rental can be inconvenient without an organized platform.

Proposed Solution

ToolShare will provide a web interface for listing tools, searching and filtering items, requesting bookings, and managing rental activity.

Objectives

- Make it easier to discover tools available for rent.
- Help owners publish listings and manage requests.
- Provide clear booking status and rental history.
- Encourage reuse of existing tools and potentially reduce unnecessary purchases.
- Deliver a responsive, documented web application for a DEPI demonstration.

Initial Scope

MVP: user accounts, tool listings, search and filters, tool details, booking requests, booking status tracking, basic profile management, and administrative review. Potential extensions, subject to time and instructor approval, include online payments, identity verification, delivery coordination, deposits, notifications, and advanced dispute resolution.

Role	Main responsibilities
Renter	Search for tools, review details, submit booking requests, and track bookings.
Tool Owner	Create and manage listings, define availability, and approve or reject requests.
Administrator	Review reports and manage inappropriate listings or user activity.
Project Team	Gather requirements, design, implement, test, document, and present the system.

Phase 1 — Project Planning & Management

1.1 Project Proposal

Project Name: ToolShare — A Peer-to-Peer Tool Rental Platform
Project Type: Web application / marketplace
Purpose: Connect people who need tools temporarily with people who own tools and are willing to rent them out.
Expected Outcome: A working MVP demonstrating listing creation, tool discovery, booking requests, and booking management.

1.2 Project Plan and Timeline

Phase	Start Date	Target Date
Project Planning & Management	2 Oct 2026	2 Oct 2026
Literature Review	9 Oct 2026	16 Oct 2026
Requirements Gathering	9 Oct 2026	16 Oct 2026
System Analysis & Design	9 Oct 2026	6 Nov 2026
Implementation (Source Code & Execution)	16 Oct 2026	30 Nov 2026
Final Presentation, Testing & Reports	16 Oct 2026	4 Dec 2026

Note: Dates follow the supplied project brief. As of 16 October 2026, the first target date has passed; confirm revised deadlines with the instructor if needed.

1.3 Team Roles and Responsibilities

Team Member	Primary Responsibility
Dareen Ahmed — UI/UX Designer	Responsible for the user flow, Figma page designs, colors, typography, UI components, and responsive design guidelines. She hands over the designs before implementation begins.
Ahmed Maged — Team Leader & Front-End Developer	Responsible for the project structure, navbar and general layout, core pages, Front-End coordination, and reviewing and merging Front-End code.
Mahmoud Sobhy — Front-End Developer	Responsible for the tool listings page, tool details page, search and filtering, and responsive layouts.
Menna Elsosy — Front-End Developer	Responsible for login and registration pages, adding and editing tools, and booking pages. She coordinates with Mahmoud Sobhy on API integration.
Mahmoud Elsayed — Back-End Developer	Responsible for database integration, APIs, authentication and permissions, and booking logic, including accepting or rejecting booking requests.

These assignments reflect the team's current plan. All members should contribute to reviews, integration, testing, documentation, and the final presentation. Coordinate dependencies early, especially between Front-End API integration and Back-End development.

1.4 Communication and Work Management

- Hold a short weekly team meeting to review completed work, blockers, and next steps.
- Use a shared task board with To Do, In Progress, In Review, and Done statuses.
- Use GitHub issues, branches, descriptive commits, and pull requests.
- Commit documentation as well as source code to the shared repository.
- Record major decisions, scope changes, and instructor feedback.

1.5 Risk Assessment

Risk	Level	Mitigation
Delayed tasks	Medium	Break work into small tasks, assign owners, and review progress weekly.
Unclear requirements	High	Agree on the MVP feature list and confirm requirements with the instructor.
Merge conflicts	Medium	Use branches, descriptive commits, pull requests, and coordinate shared-file edits.
Booking conflicts	High	Define booking status rules and validate overlapping dates in the back-end.
Insufficient test time	Medium	Test each feature as it is completed and reserve time before submission.
Team member availability	Medium	Share progress regularly and document work so others can help.
Data/security issues	High	Validate inputs, protect routes, and never commit secrets to GitHub.

1.6 Key Performance Indicators (KPIs)

KPI	Target / Success Measure
Requirements coverage	All agreed MVP requirements are implemented or their limitations are documented.
Core workflow completion	Demo supports login, tool discovery, booking request, and booking status.
Booking correctness	No conflicting confirmed bookings in the agreed test cases.
Responsive layout	Main pages are usable at mobile, tablet, and desktop sizes.
Defect tracking	Critical/high-priority bugs are fixed or documented with an agreed workaround.
Documentation completeness	Required documents and setup instructions are committed to GitHub.
Team progress	Each member's assigned tasks are reviewed at least weekly.

Phase 2 — Literature Review

This preliminary review examines existing sharing and rental platforms to identify useful patterns for ToolShare. It is a practical product review, not a claim that a formal academic systematic review has been completed. Add academic papers if required by the instructor.

2.1 Comparable Platforms

Platform	Observed model / features	Relevance to ToolShare
Peerby	Connects people with neighbors to borrow or rent items. Users can search, verify a profile, request an item, and arrange its return.	Useful comparison for peer-to-peer sharing, local discovery, booking requests, and trust.
Library of Things	Offers local rental of useful items such as drills, with online reservation and collection/return workflows.	Useful reference for item categories, availability, reservations, returns, and sustainability.
ToolShare (proposed)	Student project focused on peer-to-peer tool rental, owner listings, and renter booking requests.	Aims to combine clear listing management with searchable tools and transparent booking status.

2.2 Key Findings

- Search and discovery are essential: users need to find suitable items quickly.
- Availability and booking status should be clear to reduce confusion and conflicting requests.
- Trust matters in peer-to-peer transactions; profiles, reviews, reporting, and accurate listing details can support safer interactions.
- Collection and return expectations should be understandable, even if coordinated manually in the MVP.
- Affordability, convenience, and better use of idle tools are useful value propositions.

2.3 Identified Opportunity

ToolShare can focus on a straightforward rental journey: an owner publishes a tool, a renter searches and checks details, the renter requests dates, and the owner accepts or rejects the request. Dashboards for both roles and basic reporting can make the workflow easier to demonstrate.

2.4 Suggested Improvements

- Use consistent categories and structured listing fields such as price, location, condition, and availability.
- Show booking status and selected dates clearly to both parties.
- Include reviews and reporting only if they fit the schedule.
- Test representative scenarios for renters and owners.
- Evaluate usability, booking correctness, responsiveness, and documentation completeness.

2.5 References

Peerby. (n.d.). Peerby — Rent goods from neighbors. https://www.peerby.com/

Library of Things. (n.d.). How borrowing works. https://www.libraryofthings.co.uk/how-borrowing-works

Library of Things. (n.d.). Borrow useful things for your home, projects and adventures. https://www.libraryofthings.co.uk/

Accessed 16 October 2026. Platform features may change; verify details before final academic submission.

Phase 3 — Requirements Gathering

3.1 Scope and Assumptions

The requirements below describe a proposed MVP and should be reviewed by the team and approved by the instructor before implementation.

- ToolShare is a web application, not a native mobile application.
- Owners are responsible for listing accuracy and availability.
- Owners decide whether to accept or reject booking requests.
- The MVP may use simulated or test data for demonstration.
- Payments, identity checks, delivery, deposits, and legal dispute handling are outside the initial scope unless approved.

3.2 Functional Requirements

ID	Requirement
FR-01	The system shall allow visitors to create an account using required registration details.
FR-02	The system shall allow registered users to log in and log out securely.
FR-03	The system shall allow users to view and update profile information.
FR-04	Owners shall be able to create listings with title, description, category, rental price, location, availability, and images.
FR-05	Owners shall be able to edit, deactivate, or remove their own listings.
FR-06	The system shall display available tool listings.
FR-07	Users shall be able to search listings by keyword.
FR-08	Users shall be able to filter listings by category, price range, and location when provided.
FR-09	The system shall provide a details page for each listing.
FR-10	Renters shall be able to submit booking requests for selected dates.
FR-11	The system shall prevent conflicting confirmed bookings for the same tool and dates.
FR-12	Owners shall be able to approve or reject booking requests.
FR-13	Renters shall be able to view booking request status.
FR-14	Users shall be able to view active and past rentals.
FR-15	Users may leave a rating and review after a completed rental, subject to project scope.
FR-16	Administrators shall be able to review users/listings and handle reported content.
FR-17	The system shall validate required fields and show clear error messages.
FR-18	Users shall be able to report a listing or user for review.

3.3 Non-Functional Requirements

ID / Category	Requirement
NFR-01 Usability	Core tasks such as searching for a tool and requesting a booking should be easy to understand.
NFR-02 Responsiveness	The interface shall support common desktop, tablet, and mobile screen sizes.
NFR-03 Performance	Common pages should load within an acceptable target of 3 seconds under normal demo conditions.
NFR-04 Security	Passwords must not be stored as plain text; protected features require authentication.
NFR-05 Authorization	Users may edit only their own listings/profile; administrative actions are restricted to administrators.
NFR-06 Reliability	Booking dates and statuses must be validated to reduce duplicate or conflicting reservations.
NFR-07 Maintainability	Use clear naming, modular components, and documented setup

	instructions.
NFR-08 Compatibility	The application should work in recent versions of major browsers.
NFR-09 Data Integrity	Required fields and relationships among users, listings, and bookings must be validated.
NFR-10 Privacy	Display only the personal information needed for the rental workflow.

3.4 User Stories

User	User Story
Renter	As a renter, I want to search and filter tools so that I can find a suitable tool near me.
Renter	As a renter, I want to view tool details and availability so that I can decide whether to request it.
Renter	As a renter, I want to submit a booking request so that I can rent a tool for a chosen period.
Renter	As a renter, I want to track booking status so that I know whether the owner accepted my request.
Owner	As an owner, I want to publish and manage listings so that others can rent my tools.
Owner	As an owner, I want to accept or reject booking requests so that I control when my tools are rented.
User	As a user, I want to manage my profile so that my account information stays up to date.
Administrator	As an administrator, I want to review reported listings so that inappropriate content can be handled.

3.5 Core Use Cases

Use Case	Actor	Main Flow
Register / Log In	Visitor / User	User enters account details; system validates and provides feedback.
Publish a Tool	Tool Owner	Owner enters listing details and publishes the listing.
Find a Tool	Renter	Renter searches or filters listings and opens a details page.
Request a Booking	Renter	Renter selects dates and submits a request; system validates it and records its status.
Manage Booking	Tool Owner	Owner reviews a request and accepts or rejects it; renter sees the updated status.
Review a Report	Administrator	Administrator reviews reported content and takes an appropriate moderation action.

3.6 MVP Acceptance Criteria

- Users can register, log in, and log out.
- Owners can create, edit, and deactivate their own listings.
- Renters can search, filter, and view listing details.
- Renters can submit requests for valid dates.
- Owners can accept/reject requests and renters can view status.
- Conflicting confirmed bookings are blocked.
- Main pages are usable on mobile and desktop sizes.
- GitHub contains required documents and setup instructions.
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